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Equipment Operating & Maintenance Strategy

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Operational Excellence, Glencore Copper
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Supported by the AM Network Pty Ltd

Preface

The Asset Management Framework, Policy, Manuals and Guidelines are intended to provide leading Asset Management practices when applied and will be continually improved for the benefit of Glencore Copper.

The real value of sharing the knowledge gained from the use of these documents will come from continued discussion of the issues and rationale behind each document rather than by reading and interpreting them as individuals.

Readers are encouraged to consider these against our values around Safety, Responsibility, Simplicity, Integrity, Openness and Entrepreneurialism and how they reflect our Purpose – to responsibly source the commodities that advance everyday life. This document lays out the minimum standard for Glencore Copper and compliance to this manual will demonstrate a maturity value equal to Planned/Deployed (3) on the Glencore Copper Asset Management Maturity Scale.

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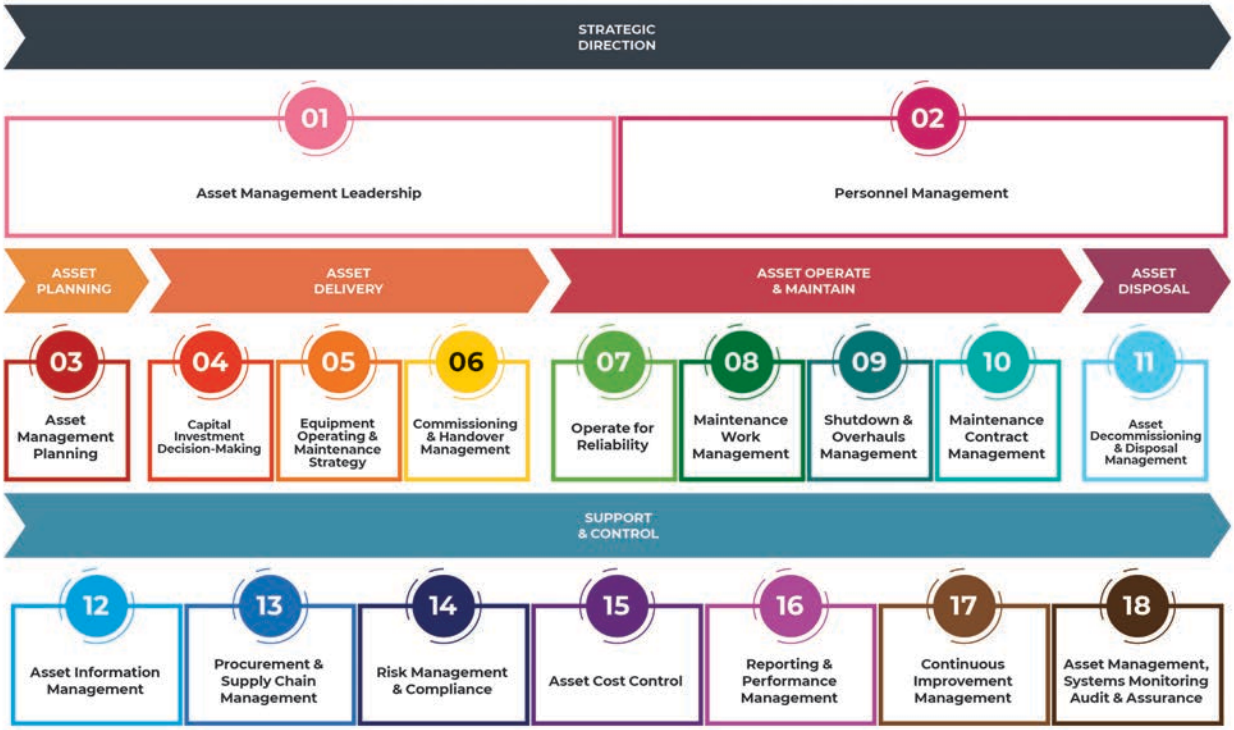
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1. Introduction

The Asset Management Policy, Manuals and Guidelines define how Glencore Copper manages the whole of life of an asset to ensure optimal value is extracted by ensuring that assets are capable of meeting business and stakeholder objectives.

Figure 1-1 describes the high-level Asset Management Framework, which enables Glencore Copper to deliver its Asset Management purpose and vision.

FIGURE 1-1 ASSET MANAGEMENT FRAMEWORK



2. Objectives

The objective of this manual for Equipment Operating and Maintenance Strategy is to define the approach, deployment, measurement of results, and continuous improvement of Equipment Operating and Maintenance Strategy in order to optimise reliability, availability, costs and risk throughout an asset's lifecycle from acquisition to disposal.

2.1 Accountabilities

Dedicated resources must be assigned, and capabilities developed within each Operation to identify work to be included in the Computerised Maintenance Management System (CMMS) and to deliver Equipment Operating and Maintenance Strategy processes that optimise reliability, availability and risk.

- Assign single point accountability for each or all of the following:
  - Equipment Operating and Maintenance Strategy identification, planning and execution
  - Equipment Operating and Maintenance Strategy procedures defined and documented
  - Ownership for the development and improvement of Equipment Operating and Maintenance Strategies
- Define and document Equipment Operating, and Maintenance Strategy competencies required

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- Plan, deliver and record Equipment Operating and Maintenance Strategy training to support the embedment of required Equipment Operating and Maintenance Strategy process competence and knowledge.
- Provide sufficient CMMS user access points to operational staff to enable the daily entry of Maintenance Work Management information into the CMMS, to assist in the analysis of Equipment Operating and Maintenance Strategy performance.

2.2 Scope

The scope includes:

- Ensure all productive assets are identified with a unique number and recorded in the CMMS asset register including the 'sort field' against each Technical Object and Equipment to match the Tag in the field.
- Determine the criticality of all productive assets down to component (maintainable item) level and recorded in the CMMS.
- Determine the functions and performance standards of all productive assets down to component level.
- Determine the operating context of all assets down to component level
- Determine tactics development methodology based on current equipment criticality. FMEA for critical assets is mandatory but sites do have the option of conducting RCM should they choose to. FMEA and PMO for low critical assets. For new or modified assets RCM shall be applied to Critical Items, FMEA for medium critical and PMO for all others.
- Determine the failure modes and their effects of all productive assets down to component level, with the means to continually update FMEAs if new failures occur or to modify FMEA's as operational data becomes available.
- Ensure Statutory, Regulatory, Codes of Practice and Health, Safety and Environment (HSE) requirements are incorporated in maintenance tactic selection and task development process.
- Develop and logically group planned maintenance tasks in the CMMS to effectively manage the known or credible failure modes.
- Define bills of material (BOMs)/parts requirements, special tools, specific task instructions and performance boundaries for all planned maintenance tasks, including logistic and access requirements.
- Develop and optimise the Asset Management Plans for all Critical productive assets with the means of benchmarking lifecycle costs.
- Utilise the Asset Management Plans to develop long term forecasts and annualised budgets (operating (OPEX) and capital (CAPEX) expenditure) with sufficient information to make risk based informed decisions on headcount, spares, and production targets.
- Measure and continually improve the effectiveness of planned maintenance tasks (both primary and secondary tasks).

2.3 Measures

Performance must be measured for each operation against the Equipment Operating and Maintenance Strategy Key Performance Indicators to ensure compliance with business processes. The foundation for measures include:

- Monitoring and reporting down to equipment level
- Committing to development of tactics for all registered equipment
- Conducting performance analysis routinely in accordance with the Equipment Operating and Maintenance Strategy Process Design and demonstrate effort on improving effectiveness and efficiency
- Conducting Equipment Operating and Maintenance Strategy process compliance evaluations
- Performing ongoing refinement activities to support and improve Equipment Operating and Maintenance Strategy discipline

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3. Definitions

Asset Register/Functional Location Structure – A list or set of related and/or hierarchical codes and associated descriptions which ultimately define and describe the individual items and/or components of equipment installed in the plant.

Asset Lifecycle Plan - Prescribed set of preventive and repair maintenance actions designed to manage the failure modes and condition of an item of equipment in a functional location.

Master Data – The core data within the enterprise that describes objects around which business is conducted. It typically changes infrequently and can include reference data that is necessary to operate the business. Master data is not transactional in nature.

Failure Mode Effect Analysis (FMEA) - is a structured approach to discovering potential failures that may exist within a product, asset, or process. Failure modes are the ways in which a process can fail. Effects are the ways that these failures can lead to waste, defects, or harm. Failure Mode and Effects Analysis is designed to identify, prioritize, and limit these failure modes.

Reliability Centred Maintenance (RCM) - provides a structured framework for analysing the functions and potential failures for a physical asset with a focus on preserving system functions, rather than preserving equipment. RCM is used to develop scheduled maintenance plans that will provide an acceptable level of operability, with an acceptable level of risk, in an efficient and cost-effective manner.

PMO – Plant Maintenance Optimisation, a process by which low critical assets maintenance tactics are determined by adopting manufacturers recommendations, adopting tactics of similar assets or the improvement of current tactic.

Asset Criticality – a process that enables the ranking of systems assets in order of business importance to ensure that effort applied is directionally proportionate to the criticality and impact to the business. Criticality is relevant for the business decision making process to manage business risk and value. Criticality is typically used to determine:

- As an input to determine the overall priority for performing a maintenance task (sometimes combined with a "Work Order Priority" entered against the specific task to give an overall priority for the task)
- To determine, at a high level, the type of risk mitigation strategy to be applied to the equipment (e.g. do condition monitoring and defect elimination on high criticality items)
- As an input into determining the optimum spare parts holdings required for the equipment item
- To provide input into the capital program so that "high criticality" equipment is given a higher priority for upgrade or replacement
- To guide reliability engineers so that they focus their reliability improvement efforts on the most "critical" equipment

Operating Context - The current condition, environment, and culture in which you operate a piece of equipment. This would include but not be limited to the following:

- Temperature (Hot, Cold or severe swings); Dirty or dusty atmosphere; Wet or dry area; Corrosive, erosive or abrasive environment; Dark or dimly lit; Noisy; Culture (Goals and expectations not clearly defined, high level of emergency/demand work); Operating outside design expectations or performance standards; Asset condition (Loose, improperly supported, improperly installed, improper design, damaged); Improper operation (Start up, shutdown, product change, setting, speed, flow, pressure); Human error (I forgot, no check lists or procedures) and organisational maturity in Asset Management

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RAM – Reliability, Availability and Maintainability (RAM) are system design attributes that can have a substantial impact on the lifecycle cost and performance of an engineered system. In RAM modelling, Reliability is the probability that an item will survive after a determined time. Availability is the percentage of time a system is considered ready to use when tasked. Maintainability is function of the failure identification time, logistics capabilities, diagnostic capabilities, tooling and instruments, repair capabilities.

Maintenance Strategy – The maintenance requirements for operating equipment in its operating context and includes:

- Preventive Maintenance – Maintenance carried out at predetermined intervals or according to prescribed criteria, aimed at reducing the failure risk or performance degradation of the equipment.
- Predictive Maintenance - Maintenance based on the equipment performance monitoring and the control of the corrective actions taken as a result
- Corrective Maintenance - Maintenance is carried out following detection of an anomaly and aimed at restoring normal operating conditions. This approach is based on the firm belief that the costs sustained for downtime and repair in case of fault are lower than the investment required for a maintenance program. This strategy may be cost-effective until catastrophic faults occur
- Reactive Maintenance – Maintenance is carried at when equipment functional failure occurs, applied to equipment that has low consequences of failure.

Inherent Risk – is an assessed level of raw or untreated risk; that is, the natural level of risk inherent in a process or activity without doing anything to reduce the likelihood or mitigate the severity of a failure.

Residual Risk – the amount of risk after inherent risk has been reduced by the implementation of risk controls

Training Needs Analysis (TNA) - a process that a business conducts in order to determine all the training that needs to be completed in a certain period to allow their team to complete their job as effectively as possible, as well as to progress and grow.

Standard Job/SWI/Corrective Task/Corrective Procedure – all of these items represent a fully planned and costed corrective job that resides in the CMMS includes, job procedure, spares, specialist equipment, consumables and resources required to execute the job.

4. Approach

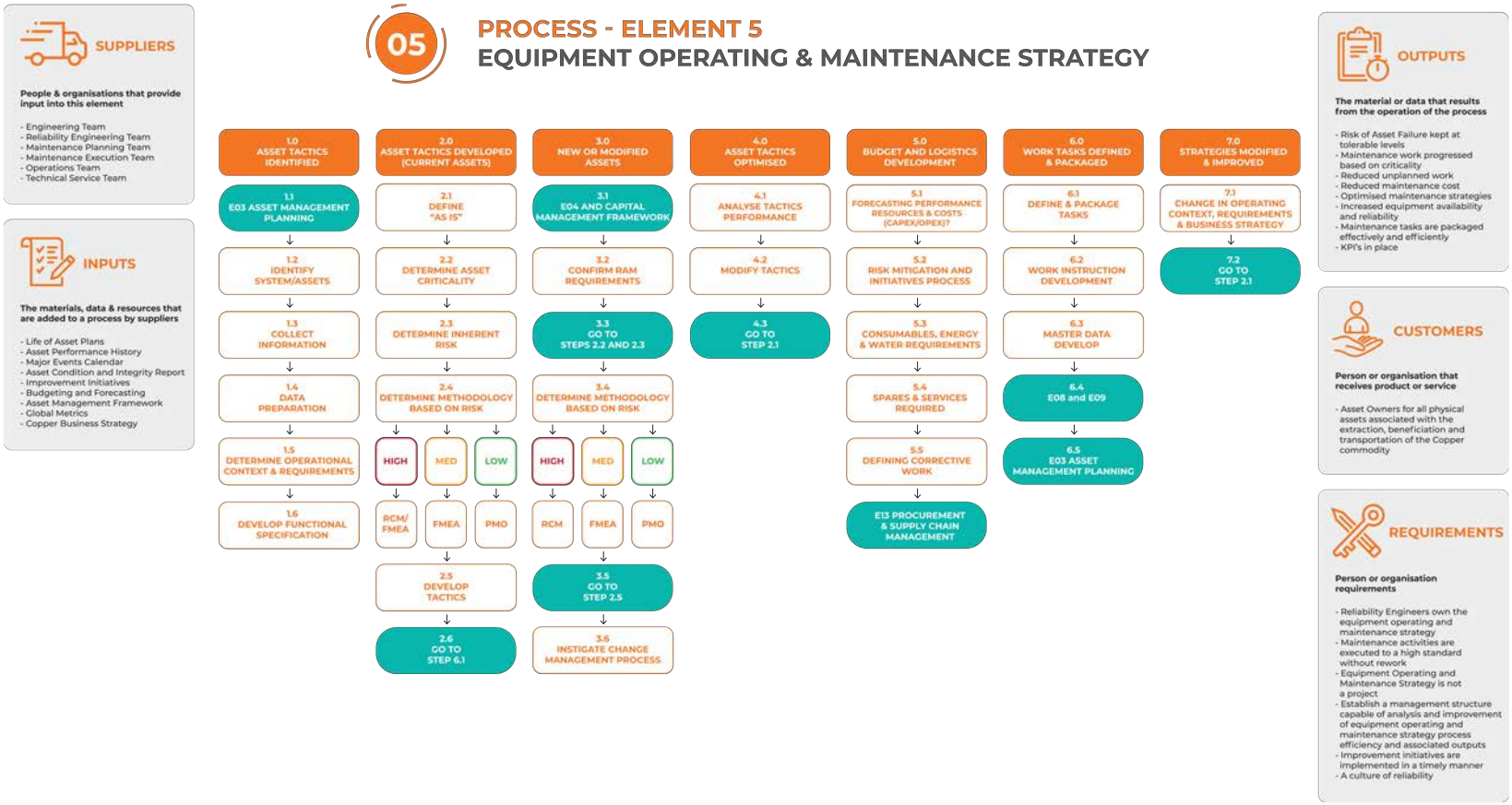
The approach for Equipment Operating and Maintenance Strategy is described through:

- Process Flow - pictorially illustrates the key Equipment Operating and Maintenance Strategy activities, refer to Figure 2. The Equipment Operating and Maintenance Strategy Process is somewhat shaped by the inherent processes upon which the CMMS Maintenance module is designed and built. The intent of the workflow is to ensure that maximum effort is applied to the systems and assets that are of the most value to Glencore Copper resulting in a rapid and cost-effective deployment strategy
- Process Description – defines in more detail the key Equipment Operating and Maintenance Strategy activities, refer to Section 4.2
- Process RACI Matrix - indicates for each activity, which Function undertakes the activity and which Functions are, Accountable, which are Consulted and Informed prior to final decision/action, and those which are Informed of any outcomes and actions, refer to Section 4.3

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4.1 Process Flow

FIGURE 2-1 - EQUIPMENT OPERATING AND MAINTENANCE STRATEGY FLOW



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4.2 Process Description

TABLE 4-1 PROCESS DESCRIPTION

PROCESS STEP	DESCRIPTION
1.0 Asset Tactics Identified	**Assets have management processes and activities in place, determining the operating and maintenance strategies to support the Asset Management objectives and goals. Tactics prioritised on Criticality**
1.1 E03 Asset Management Planning	Consult with Element E03 Asset Management Planning and associated Asset Management Plans to understand the maintenance strategy and requirements prior to system/asset selection. Ensure that business cases are developed and approved in line with Glencore Copper requirements.
1.2 Identify System/Assets	Identify the System / Asset that require Equipment Operating and Maintenance Strategy. This can come from business needs such as risk reviews, incident investigations, asset acquisition, modification, and data analysis.
1.3 Collect Information	There are various inputs that can lead to the development or refinement of a Maintenance Strategy. These include: • Procurement of new asset • Modification of existing asset • Asset failure (lost production) • Business need to optimise maintenance strategies • Change in Operating Planning • Change in Life of Asset Planning This process step requires the responsible person to assess why there is a need and collect information to support the need. Information to be collected could include: • CMMS Master Data • CMMS cost and failure transactional data • KPI reports • Down time reports • RCM/FMECA data for similar asset • OEM manuals • Other reliability data, i.e. RAM, Weibull Analysis, Failure Rates etc. • Asset data sheets • Drawings • Standards, legal, regulatory, statutory and Safety Instrumented Systems (SIS) • Business Plans • Operating Plans • Life of Asset Plans (Asset Management Plans)
1.4 Data Preparation	All necessary data shall be collated and put into a format that is readily available and is useable.
1.5 Determine Operational Context and Requirements	It is important to understand the context in which the asset is required to operate, this will ensure that maintenance tactics are developed or modified to take into

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	consideration this operational context. Things to consider when determining operating context include: • Environmental: Temperature (hot, cold, or severe swings); Dirty or dusty atmosphere; Extremely wet or dry; Corrosive, erosive or abrasive environment; Dark or dimly lit; Noisy • Cultural: Goals and expectations not clearly defined; High level of emergency/demand work; Assets operated outside design expectations or performance standards; Improper operation (start up, shutdown, product change, setting, speed, flow, pressure); Systems induced human error (I forgot, no check lists or procedures) • Asset condition: Mechanically (improper design, improperly installed, loose, imbalance, contaminated), electrically (poor connections and protections), and structurally (corroded, movement) • The primary purpose of the item and how it contributes to the business • Whether the item is part of a batch or flow process • If redundancy exists • Safety standards applying to the asset • Environmental standards • Typical duty and operating cycles • Typical maintenance and support arrangements • Whether stockpiles exist which allow plant down time without affecting overall production • Impact of repair time on production • The level of spares carried. • Logistic and access requirements for conducting maintenance activity • Asset Management Maturity of the organisation For example, the operation of gearboxes in different geographical locations where the temperature variation is significant, meaning the maintenance tactics would need to be adjusted to take in consideration that geographical location. Consideration also needs to be given to Operational requirements in that a change of Operational strategy i.e. production increase will have a profound effect on asset maintenance requirements as will LOM changes.
1.6 Develop Functional Specification	A comprehensive functional specification that clearly defines the functional, performance, reliability, maintainability and useful life or remaining life of an asset system and asset. Both Primary and Secondary functions shall be identified.
2.0 Asset Tactics Development	**Asset Tactics are developed using a team of Reliability and Subject Matter experts from cross functional areas**
2.1 Define “as is”	There needs to be a means by which success can be measured, this involves taking a snapshot of the ‘As Is” and then comparing to the “To Be” to understand the potential business benefit. Things to consider when determining the “As Is” include: • Mean Time Between Failure (MTBF) • Availability • Operational cost, inclusive of lost opportunity cost • Maintenance cost

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	• Failure modes • Maintenance task frequency and duration • Resource requirements • Spares requirements • Risk • Operator Maintenance (refer to Element E07 Operate for Reliability)
2.2 Determine Asset Criticality	Asset Criticality is a process that enables the ranking of systems and assets in order of business importance to ensure that effort applied is appropriately proportionate to the value of the asset to the organisation. In the context of Equipment Operating Maintenance Strategy Development this ensures: • Determination of the overall priority for performing a maintenance task (sometimes combined with a "Work Order Priority" entered against the specific task to give an overall priority for the task) • Determination, at a high level, the type of risk mitigation strategy to be applied to the equipment (e.g. do condition monitoring and defect elimination on high criticality items) • Determination of the optimum spare parts holdings required for the equipment item • Guidance to reliability engineers so that they focus their reliability improvement efforts on the most "critical" equipment The first step in the process requires an analysis of the system/s to determine its criticality. Once this is achieved then further analysis can be conducted to determine the asset criticality. The process used to determine asset criticality shall be quantitative, and consideration for qualitative assessment only when quantitative is not possible.
2.3 Determine Inherent Risk	To understand the full impact an undesirable event would have at Glencore Copper a determination of the Inherent Risk needs to be conducted. This should be done using the Glencore Copper corporate HSEC standards and shall be modified to take into consideration risk criteria at each operation. Element E14 Risk Management and Compliance shall also be consulted during this process step. The Glencore Copper Risk Matrix in which risk is classified three (3) categories, being High, Medium and Low.
2.4 Determine Methodology Based on Risk	The introduction of the E05 manual at Glencore Copper is through a phased approach. Given the onerous nature of Reliability Centred Maintenance (RCM) it deemed that this methodology will only apply to new or highly modified critical equipment. To ensure that the amount of effort applied to develop equipment operating and maintenance strategies is proportional to the risk exposure to Glencore Copper a risk-based approached is used. Various risk management processes can be used depending on the level of risk. These processes include:

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Reliability Centred Maintenance (RCM) - RCM is a structured process which sequentially asks the following seven questions about the asset or system under review:

- Functions - what are the functions and associated performance standards of the asset in its present operating context?
- Functional Failures - in what ways does it fail to fulfil its functions?
- Failure Modes - what causes each functional failure?
- Failure Effects - what happens when each failure occurs?
- Failure Consequences - in what way does each failure matter?
- Proactive Tasks - what can be done to predict or prevent each failure?
- Default Actions - what should be done if a suitable proactive task cannot be found?

Should be applied to High- Risk assets as this will yield an optimal outcome regarding asset risk, cost, and performance. Only to be applied for New or highly modified assets. See process step 3.3
The minimum standard for current systems and assets is FMEA but sites do have the option of RCM should they wish to do so.

Failure Modes and Effect Analysis (FMEA) - is used to identify the ways in which components, systems or processes can fail to fulfil their design intent and identifies:

- All potential failure modes of the various parts of a system,
- The effects these failures may have on the system,
- The mechanisms of failure, and
- How to avoid the failures, and/or mitigate the effects of the failures on the system through the development and execution of maintenance tactics

It is a less onerous process than RCM but yields good results due to the structured and repeatable process.
Should be applied to High and Medium Risk Assets as this will provide significant value for effort application.

Plant Maintenance Optimisation (PMO) - PMO differs in methodology to both RCM and FMEA. PMO is a methodology that can quickly improve current maintenance programs or implement new programs for asset of a low criticality ranking. When applied to current maintenance programs, PMO can quickly and effectively optimise maintenance programs currently undertaken. It does so by applying logical and risk-based decision making with respect to**:**

- Task value add – does the task address a failure mode/ mechanism
- Task type – can we move from a preventive to a predictive strategy
- Task frequency – is there sufficient evidence to support a change in the task frequency
- Task duration – does the allocated duration of the task reflect the real -world actual duration
- Is there sufficient evidence to support a Run to Failure strategy?

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	Should be applied to Low Risk Assets and will provide significant gain regarding the cost of conducting maintenance tasks. Each of the three processes detailed above will define what are the maintenance tasks, these will require further analysis. Unless an asset experiences definitive wear out characteristic failure modes then Maintenance tasks should be predictive in nature and condition based maintenance tactics developed. Otherwise, schedule restoration or schedule discard strategy shall be adopted
2.5 Develop Tactics	The following provides a guide to the development of maintenance tactics: • Identify team member and resources, these members form part of the working group and shall conduct workshops to enable the developed tactics to be challenged before implementation • Determine tactics development methodology based on asset criticality • Review functional specification of process, systems, and assets • Review all data collected and consolidated • Consider how the tactics are going to be developed, for instance, in a spreadsheet or proprietary software • Select an appropriate strategy for the management of each failure mode and develop maintenance tasks to mitigate each failure mode. Each tactic shall include Primary Task (PM02), Corrective Task (PM01), Breakdown task (PM03). Primary Task detail should be written to a level that will enable a tradesperson to conduct the task safely and to the desired quality. Corrective and Breakdown tasks shall be of sufficient detail to provide estimates on cost, duration, spares, and frequency, and shall be written with enough detail to identify what work needs to be done and not necessarily how to do that work • Analyse spares needs. Following determination of their maintenance requirements, all Critical assets shall undergo a critical spares review. The primary aims will be to reduce the risk of stock-outs impacting on business performance and, to minimise the cost of holding unnecessary stock. • The review process will be conducted by maintenance, engineering, and operations personnel together and will address: ○ How the maintenance requirements of the asset impact on spares requirements. A Level of Repair Analysis (LORA) should also be conducted to determine whether spares will be repaired or replaced and if so, by whom ○ What the consequences are of not holding a spare ○ Whether the spares requirement can be anticipated (“dependent demand”) ○ What stock holding is needed if the requirement for a spare cannot be anticipated (“independent demand”). • Conduct challenge workshops using a team of cross functional Subject Matter Experts (SME’s) to review the tactics and make necessary adjustment. The team should consist of: ○ Facilitator: The Facilitator is a highly trained expert in application of Reliability-Centred Maintenance, FMEA and PMO who guides the analysis team through the validation of the tactics

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	○ Operations and Maintenance Supervisors, who can provide SME assistance for the area under consideration. ○ Operations and Maintenance representatives: It is not possible for maintenance personnel to answer the structured questions without involvement of operations ○ Specialists: Input from specialists such as process experts • Conduct Risk Assessment to determine Residual Risk • Determine if manpower requirements are within current operational capacity • Sign off on final tactics Typically, Maintenance Strategies to be considered include: • **Condition-Based Maintenance (CBM)** - a strategy for evaluating equipment performance to determine the operating condition of the asset enabling early detection of developing defects and performance variation, allowing personnel to then make informed maintenance decisions with an improved understanding of maintenance activity scope and timing. • **Fixed Time or Preventative Maintenance (FTM) –** a strategy directly linked to equipment that has age-related failures. Age-related failures are symptomatic of equipment that has wear-out characteristics where, upon reaching a certain point, there is a rapid increase in the probability of a failure. These failures tend to be associated with fatigue, corrosion, usage, and direct wear conditions. • **Run to Failure –** a strategy where Condition Based Maintenance or Preventative Maintenance tasks cannot be justified on cost, health, safety, environmental, quality, or operational benefits, the decision must then be to run to failure. When Preventative Maintenance and Condition Based Maintenance strategies are not applicable and the risk of running to failure is unacceptable then redesign must be done. • **Fault Finding Tasks –** a strategy used if the failure mode is hidden. A hidden failure is one that on its own will not become evident under normal operating conditions. While hidden failures do not have any direct consequences, they may expose the operation to the risk of multiple or significant failures. Examples of hidden failures include the non-operability of the standby pumps, pressure relief switches, crane anti-collision protection, back-up generators, pull wires, fire alarms, etc. • **Redesign –** when all the above strategies are not appropriate then redesign shall be considered.
3.0 New or Modified Assets	**The New Asset and Modification sub-process describes the steps that are required to ensure successful and rapid development and implementation of acquiring new assets or modifying existing assets.**
3.1 E04 Capital Management Decision-Making	The Glencore Copper Capital Management Framework and Asset Management Framework element E04 shall be consulted as part of this process step.
3.2 Confirm RAM Requirements	Reliability, Availability and Maintainability (RAM) modelling is mandatory in the pre-feasibility phase of the asset lifecycle according to Glencore Copper Study Guidelines 044-CMF-GCU-GDL-0018, 020-CMF-GCU-GDL-0002, 021-CMF-GCU-GDL-0003. The

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	RAM requirements should be confirmed to ensure that the maintenance tactics that are developed are capable of achieving RAM targets
3.3 Go to Steps 2.2 and 2.3	Step 2.2 Determine Asset Criticality and Step 2.3 Determine Inherent Risk
3.4 Determine Methodology Based on Risk	See process step 2.4
3.5 Go to Step 2.5	To ensure that the new acquired asset meets all RAM requirements the asset has to be registered in the CMMS Technical Hierarchy and master data created including maintenance tactics data. Therefore, following steps in Process step 2.5 will ensure that tactics are created for new assets and adjusted for modified assets.
3.6 Instigate Change Management Process	Follow the Glencore Copper Change Management process to have the change authorised and finalised
4.0 Asset Tactics Optimised	**Periodically Asset Tactics shall be analysed and optimised**
4.1 Analyse Tactics Performance	It is essential that asset tactics performance is analysed to ensure that Glencore Copper are achieving the maximum value and achieving the desired balance or Risk, Cost and Performance. There are several mechanisms that could trigger a review of Asset Tactics including: • Annually for Critical Assets • Actual maintenance costs exceed budgeted costs • Post unacceptable failure event • As legislation changes • As standards change • As business objectives change • As technology changes • If assets become obsolete • Adherence to maintenance schedule is regularly non-compliant • 3C – Concern, Cause, Countermeasure Process • 5C – Concern, Cause, Containment, Countermeasure, Culture Process • RCA • Incident Cause Analysis Method (ICAM) Tactics performance to analyse include: • Duration – is the job taking longer or less time than what the original analysis recommended • Frequency – is this correct. The intent of predictive tactics is to find impending failure before they cause functional failure, therefore, corrective work should be required. Analyse the amount of subsequent corrective work that is generated from the PM02. • Resource Requirements – are there enough trades allocated the tasks, materials correct, and any specialist equipment or tools correct • Cost – is the actual cost of the work within 10% of the planned cost • Asset Availability – are KPI's for asset availability within limits • Reliability – is the asset within reliability targets when utilised

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	• Have all failure modes been identified and addressed? • Are tactics packaged to provide efficient and effective routing? • Risk – has the risk threshold for the asset changed? • If significant changes have occurred, then tactics shall be modified.
4.2 Modify Tactics	Tactics shall be modified as per Process step 2 of this manual
5.0 Budget and Logistics Development	**Defining budgets and the required logistics to support the operation and maintenance strategy, using a level of detail based on equipment criticality. Element E15 Asset Cost Control shall be consulted for this process section.**
5.1 Forecasting performance, resources and costs (CAPEX/OPEX)?	The Glencore Copper annual business plan determines the production requirements for the period and therefore the operational and maintenance strategy should be aligned to the requirements of that plan. Risk, Cost and Asset Performance shall be forecasted to ensure KPI's are met. Costs to be forecast include: • Fixed – costs derived for PM02's and forecast operating expenditure • Corrective – derived for the expected (predicted) planned corrective costs • Breakdown – derived for the expected level of reactive work • CAPEX – derived from the capital investment program that is required to meet the business plan
5.2 Risk mitigation and initiatives process	Risk Management is defined in a number of other Elements within the Glencore Copper AM Framework, as well as the Glencore Copper Risk Management Framework. The reader needs to be aware of and have an understanding of the broader requirements regarding risk management and should consult with the requirements of this manual, Element E04 Capital Investment Decision Making, Element 03 Asset Management Planning, Element 14 Risk Management and Compliance, and the Glencore Copper Risk Management Framework.
5.3 Consumables, energy and water requirements?	Cost forecasts are required for: • Consumables • Energy • Water
5.4 Spares and services required	Estimates of spare parts developed during process step 2.5 shall be costed and the requirements of E13 Procurement and Supply Chain Management consulted. Any services that are required to support the Equipment and Maintenance Strategy shall be costed, together with any specialist tools and equipment.
5.5 Defining Corrective Work	Estimates of corrective work shall be made for each predictive maintenance strategy, this should include frequency of occurrence, costs, spare parts, work duration and resources required. Where sufficient information is available standard jobs can be set up in the CMMS as defined by process step 6.3. For example, Maintenance Tactics have been developed for a gearbox and executing the tactics will provide information on the health of that gearbox, this information can be used to estimate the remaining useful life of the gearbox. Once this is known, then for budgeting purposes we can allocate capital to replace the gearbox at predicted end of life. We can estimate the cost of the replacement gearbox, how long it will take to replace, develop a standard job for the replacement and then follow the

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	requirements of Element 04 Capital Investment Decision Making, provide input into Element 03 Asset Management Planning and Element 14 Risk Management and Compliance
6.0 Work Task Defined and Packaged	**Tasks shall be grouped into logical, effective and efficient packages of work**
6.1 Define and Package Tasks	Refer to Glencore Copper Master Data guideline which provides information regarding the relationships between Master Data elements. Once tasks have been authorised, they need to be grouped into logical and practical packages of work. There are many options available to do this, and it is important that the person/s undertaking this task are familiar with how the data is to be implemented into the CMMS and the functionality of the CMMS. When considering task packaging, tasks should be package by: • System/process area • Functional Location/Equipment • Activity Type • Resources type (e.g. Mechanical) • System Status (e.g. Running) • Task Frequency (e.g. Monthly) • Routing (e.g. Condition Monitoring, Lubrication) If frequencies must be adjusted in the packaging process, it would be very risky to increase frequencies for critical assets and for fault finding tasks, but for other assets some adjustment to the frequency would be acceptable On completion of task packaging, test the routine and maintenance tasks in the field, adjust as necessary.
6.2 Work Instruction Development	Once task have been packaged, creation of work instructions shall be undertaken. The standard Glencore Copper work instruction template shall be used for this purpose. The work instruction is then implemented into the CMMS in the form of a master document.
6.3 Master Data Development	Follow Glencore Copper Master Data process for the development or modification of the CMMS Master Data
6.3 Go to E08 Work Management Standard	At this point in the process there is a direct tie in with manual E08 Maintenance Work Management where created or modified system generated work orders follow the work management process.
7.0 Strategies Modified and Improved	**The process of systematically improving, and document equipment operating and maintenance strategies when there is a change in demand or operating context**
7.1 Change in Operating Context, Requirements and Business Strategy	Equipment Operating and Maintenance Strategies shall be modified and/or improve when the following occurs: • Setting review periods for implementation, based on equipment criticality. Annually for Critical Assets

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	• Changes in legislative requirements and engineering practice standards. • Receiving alerts or improvements. • Investigating identified flaws in the asset management strategy. • Incorporating significant changes when the operating context occur. • Changes to mine and production plans New strategies or modified strategies must trigger a new TOC (total ownership cost) review.
7.2 Go to Step 2.1	Go to process step 2.1 and follow the strategy development process

4.3 Process RACI Description

The RACI matrix below in Table 2 addresses the high-level functional elements of the Equipment Operating and Maintenance Strategy process. For the overall Asset Management RACI matrix, refer to the Asset Management Intranet page.

Note: The HSE and Commercial Departments do not appear in this high-level RACI, although their involvement will be required from time to time dependent on the nature of the variance.

The codes used in the RACI analysis are defined as follows:

TABLE4- 2 RACI

R	Responsible	The function(s) which will perform an activity or directs others to perform the activity.
A	Accountable	The function which ultimately “owns” the activity – there should only be one function
C	Consulted	The function that needs to feedback and contribute to the activity
I	Inform	The functions(s) that needs to be informed after a decision or action is taken if relevant. (via 1-way communication)

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Process Step \ Functional Role	Maintenance Manager/Supt	Operations Supervisor/ Operator	Maintenance Supervisor/ Trades	Supply/ Warehouse	Maintenance Planner	Reliability Superintendent	Reliability Engineering	Project Engineering
1.1 E03 Asset Management Planning	A					R		
1.2 Identify Systems/Assets	R				C	A		
1.3 Collect Information	C	R	R		R	A		
1.4 Data Preparation	C		R		R	A		
1.5 Determine Operational Context and Requirements	C	R	C			A	R	
1.6 Develop Functional Specification	C					A	R	
2.0 Asset Tactics Development								
2.1 Define “as is”	C		C		R	A	R	
2.2 Determine Asset Criticality	C		C		C	A	R	
2.3 Determine Inherent Risk	C	R	R		R	A	R	
2.4 Determine Methodology Based on Risk	I		C		C	A	R	
2.5 Develop Tactics	C	R	R		R	A	R	
2.6 Go to Step 6.1	I					A	R	
3.0 New or Modified Assets								
3.1 E04 Capital Investment Decision Making and Capital Management Framework	A						R	R
3.2 Confirm RAM Requirements	C		C		C	A	R	
3.3 Go to Steps 2.2 and 2.3	I	R	R		R	A	R	
3.4 Determine Methodology Based on Risk	C		C		C	A	R	
3.5 Go to Step 2.5	I	R	R		R	A	R	
3.6 Instigate Change Management Process	A		R		R	R	R	
4.0 Asset Tactics Optimised								
4.1 Analyse Tactics Performance	C	R	R		R	A	R	
4.2 Modify Tactics	I	R	R		R	A	R	
4.3 Go to Step 2.1	C					A	R	
5.0 Budget and Logistics Development								
5.1 Forecasting performance, resources and costs (CAPEX/OPEX)?	R		C		C	A	C	
5.2 Risk Mitigation and Initiative Processes	C		C		C	A	R	

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Process Step \ Functional Role	Maintenance Manager/Supt	Operations Supervisor/ Operator	Maintenance Supervisor/ Trades	Supply/ Warehouse	Maintenance Planner	Reliability Superintendent	Reliability Engineering	Project Engineering
5.3 Consumables, Energy and Water Requirements	C	R	C		R	A	R	
5.4 Spares and Services Required	A			C		A	R	
5.5 Defining Corrective Work	I		R		R	R	C	
6.0 Work Tasks Defined and Packaged								
6.1 Define and Package Tasks	I		R		R	A	R	
6.2 Work Instruction Development	I		R		C	A	C	
6.3 Master Data Development	I		C		R	A	R	
6.4 E08 Maintenance Work Management and E09 Shutdown and Overhaul Management	I				R	A	R	
7.0 Strategies Modified and Improved								
7.1 Change in Operating Context, Requirements and Business Strategy	A					R	C	
7.2 Go to Step 2.1	I					A	R	

TABLE 4-3 FUNCTIONAL DESCRIPTIONS

Function	Description
Maintenance Manager/Superintendent	Provides leadership in the short to medium term time horizon. Responsible for establishing the maintenance plan, budget and managing the execution of maintenance activities. Ensure that the maintenance strategy is fit for purpose and participates in the development of the strategy and tactics
Operations Supervisor/Operator	Responsible for ensuring that assets are operated within design limits and are done so safely. Provides input into the maintenance strategy development process as required.
Maintenance Supervisor/Trades	Responsible for carrying out planned and non-planned maintenance activities. Assists in the development of the maintenance tactics
Supply/Warehouse	Responsible for providing input into the maintenance strategy development process regarding the identification of spares parts for maintenance tactics
Maintenance Planner	Responsible for planning the maintenance tactics and is responsible for some areas of the maintenance strategy development process.

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Reliability Superintendent	Accountable for developing the site Reliability Framework including the deployment of the corporate Equipment Operating and Maintenance Strategy Manual. Also accountable for the development and deployment of the site-specific Reliability processes and procedures
Reliability Engineer	Responsible for the deployment of the corporate Equipment Operating and Maintenance Strategy Manual and the development and deployment of the site-specific Reliability processes and procedures.
Project Engineer	Responsible for providing technical input regarding the acquisition of new assets or major modification of existing assets, and subsequent maintenance strategies and tactics developed.

5. Deployment

The deployment of the Equipment Operating and Maintenance Strategy approach across all business areas is through:

- Awareness presentation
- Position description alignment
- Competency assessment
- Training, coaching, and mentoring

5.1 Awareness Presentation

This Manual is supported by a slide presentation and video to help facilitate communication and to raise awareness on how it contributes to the Asset Management Policy and business success.

5.2 Position Description Alignment

To effectively deploy this Manual, the essential process steps and activities need to be diligently mapped and assigned to approved positions within Glencore Copper.

The RACI matrix assists in the visualisation that everyone is working towards the same goal. It clarifies, not only the individual's responsibilities, but also that of others. It has proven to be an effective tool to quickly understand and close the gaps between current and desired practices and business process activities.

Position descriptions including KPIs need to be aligned to the RACI matrix and measures of success. This is the basis for ongoing performance reviews and the creation and execution of individual development plans.

5.3 Competency Assessment

This Manual is supported by an Asset Management competencies assessment process and each site is responsible for assessing Asset Management competencies which is used to:

- Guide recruitment, selection, development, and deployment of people in asset management roles across the company

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- Provide employees responsible for managing assets with feedback, development, coaching and recognition for their performance
- Map, build and manage the asset management talent pool across the company
- Identify strengths and development needs in asset management practitioners

5.4 Training, Coaching and Mentoring

Coaching and mentoring is another way to assist employees improve their understanding of their role and to improve performance.

Coaching is generally about improving performance in the short term where mentoring is about achieving high performance and career aspirations over a longer term.

Coaching or mentoring requires addressing that a person is not currently up to the required level to perform their role as effectively and efficiently as they could. The build-up of skills to perform efficiently and effectively requires a large investment of time and cooperation.

Coaches and Mentors need to be carefully selected and must be committed for the long term to effect maximum success.

For this manual, the following training shall be conducted as a minimum:

- RCM
- FMEA
- PMO
- Criticality Analysis

Each operation is responsible for conducting a Training Needs Analysis (TNA) to determine which functions are to undergo training. Element 07 Operate for Reliability shall also be consulted for training needs.

6. Measures of Success

The following guiding principals have been used for the development of metrics:

1. An Input – Output model that measures key process and equipment performance parameters
2. Metrics are at a global level and focussed on input and output performance rather than lower level metrics
3. Less metrics are preferred with no more than 4 input and output metrics at the global level
4. Need to be able to measure from data that comes from the CMMS, Operational and Production Systems
5. Metrics need to be targeted to drive long term discipline

Refer to Glencore Copper's AM Manual E16 Reporting and Performance Management Manual for current KPI's

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TABLE 6-1 MEASURE OF SUCCESS

Process Input Metrics

Metric	Criticality
Definition	**Asset Criticality** – a process that enables the ranking of systems and assets in order of business importance to ensure that effort applied is directionally proportionate to the value of the asset to the organisation.
Performance Indicator	100% of all Assets in the CMMS have criticality (A,B,C or 1,2,3) assigned
Calculation	Number of Assets in CMMS/Number of Assets with Criticality Assigned
Example	ABC indicator populated on all SAP technical objects (FLOC)
Rational	Criticality is typically used to determine: • As an input to determine the overall priority for performing a maintenance task (sometimes combined with a "Work Order Priority" entered against the specific task to give an overall priority for the task) • To determine, at a high level, the type of risk mitigation strategy to be applied to the equipment (e.g. do condition monitoring and defect elimination on high criticality items) • As an input into determining the optimum spare parts holdings required for the equipment item • To provide input into the capital program so that "high criticality" equipment is given a higher priority for upgrade or replacement

Metric	Updated Maintenance Tactics
Definition	A measure of the number of Maintenance Tactics that are updated
Performance Indicator	Increase in the Number of Maintenance Tactics updated annually and reported monthly
Calculation	Number of Maintenance Tactics Updated/Number of Maintenance Tactics
Rationale	Continuous Improvement shall be a Business as Usual activity and as such, Maintenance Tactics shall be reviewed on a regular basis to ensure optimisation of asset Risk, Cost and Production performance occurs.

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Process Output Metrics

Metric	Labour Hours Executing PM's
Definition	A measure of the number of labour hours executing PM's compared to the total labour capacity available
Performance Indicator	~50% PM Labour Hours
Calculation	Total Number PM Labour Hours/ Labour Capacity
Example	Number of Hours for PM01, PM02 and PM03 in SAP
Rationale	PM's provide the minimum level of maintenance required to ensure plant performance is within desired parameters and therefore should form the majority of work undertaken from a labour hour's perspective. Industry widely accepts that for a mature Asset Management organisation PM's should make up in the order of 50% of the total work planned and scheduled for the period.

Metric	Corrective Work
Definition	A measure of the number of planned corrective tasks that are generated through the execution of Number of Primary Preventive/Predictive Tasks PM's
Performance Indicator	1 Corrective Task is generated as a result of performing 6 PM tasks on a piece of equipment
Calculation	Number of Corrective Tasks/Number of Primary Preventive/Predictive Tasks (PM's)
Example	Number of PM01's generated as a result executing PM02's in SAP. PM01's are generated as subsequent work orders to the PM02
Rationale	The main intent of PM's is to predict, prevent or significantly reduce the consequences of functional failure of equipment. PM's should therefore provide the ability to detect these events before they occur and as such it is expected that planned corrective work will be identified on average after undertaking 6 PM tasks. This ensures that the frequency of doing PM's balances risk, cost and equipment performance.

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Metric	Primary Preventive/Predictive Tasks (PM's) Costs
Definition	A measure of the cost of PM's before the implementation of this manual or the cost pre-PM optimisation versus the cost after implementation and optimisation
Performance Indicator	Reduction in PM cost Annually reported Quarterly
Calculation	PM Cost Current Financial Year/PM Cost Previous Financial Year
Example	Cost of PM02's in SAP
Rationale	Provides a means by which PM costs can be measured with the intent on seeing an improvement year on year. Optimisation of PM's should see a reduction in Emergency/Breakdown work and thus overall maintenance cost.

Metric	MTBF
Definition	Mean Time Between Failure quantifies asset Reliability and represents the average Utilised Time (UT) between Unscheduled Maintenance (UM) events
Performance Indicator	Increase in MTBF Annually reported Quarterly
Calculation	Utilised Time (UT)/# Unscheduled Maintenance Events (UM)
Example	Utilisation time taken from delay accounting systems and the number of unscheduled maintenance events (PM03 typically) from SAP
Rationale	Provides a means by which equipment reliability is measured. An increase in MTBF indicates that failure frequency is reducing reliability is improving and is directly related to equipment Availability improvement.

Metric	MTTR
Definition	Mean Time To Repair is the average time taken to repair equipment after unscheduled maintenance events
Performance Indicator	Reduction in MTTR annually reported Quarterly
Calculation	Unscheduled Maintenance (UM) Time (Hrs)/# Unscheduled Maintenance (UM) Events
Example	Utilisation time taken from delay accounting systems and unscheduled maintenance hours (PM03 typically) from SAP
Rationale	Reducing the time it takes to recover from an unscheduled maintenance event has a direct input in the increase in equipment availability by reducing the time equipment is not producing

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Metric	Primary Preventive/Predictive Tasks (PM's) Compliance
Definition	Measures how many PM's have been complete in the planning period.
Performance Indicator	100% of all PM's Complete. Must have time confirmed and the work order is maintenance complete
Calculation	PM's Complete/PM's Scheduled
Example	PM's time confirmed and MCMP in SAP where MCMP is Maintenance Complete
Rationale	PM's form the foundation of the maintenance programme and must be complied to, as compliance ensures equipment Reliability and Availability.

Metric	Availability
Definition	Percentage of Calendar Time an asset or plant was physically available to be used
Performance Indicator	Site Specific Target in line with Life of Asset Plan
Calculation	Available Time (AT)/Calendar Time (CT)
Rationale	Availability is the measure of time loss and is directly related to the Reliability of an asset or system and is the probability that an asset or system will be available to perform its function when called upon to do so. A high availability is required to ensure that Glencore Copper meets its production targets.

7. Improvement

The process to review and continually improve the approach and deployment for Equipment Operating and Maintenance Strategy is scheduled annually, or when there is a sudden change in the Operating Context. As the organisation matures a longer review period could be considered. Criticality shall be reviewed annually, and strategies modified accordingly.

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8. References

The following documents provide detailed support material that should be read in conjunction with this Manual to provide a comprehensive understanding of Equipment Operating and Maintenance Strategy.

TABLE 8-4 REFERENCES

DOCUMENT TITLE	DESCRIPTION
Asset Management Policy	This document defines the overall direction of asset management across Glencore Copper.
Asset Management Plans	These documents define how each system, sub-system and assets are to be managed within the timeframe that the plan has been developed. Amongst things it lays out the maintenance strategy requirements for Glencore Copper assets.
Asset Management – Framework	This document provides the overarching Framework that is required of Glencore Copper in the management of its assets globally

9. Revision History

VERSION NO.	REASON FOR REVISION	DATE
0.1	New Manual to support the Asset Management Policy	08/10/2020
0.2	Gather Documents and Create Draft Manual	15/10/2020
0.3	Review and Signoff 1st Draft Manual	06/11/2020
0.4	Review by Site AM reps and comment	11/12/2020
1.0	Update Site Signoff and Publish	29/01/2021

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10. Appendix A

TABLE 5: STAKEHOLDER SIGNOFF

Stakeholder Name	Site or Department	Contact	Stakeholder Signoff
Rohan Reisener	Operational Excellence - Australia	rohan.reisener@glencore.com.au	DocuSigned by: Rohan Reisener 9B7E3453600948D...
Peter Zuliani	Operational Excellence Metallurgical Assets	Peter.Zuliani@glencore.ca	DocuSigned by: Peter Zuliani 447083DDCCF34B8...
Dino Moll	Operational Excellence – South America	Dino.Moll@glencore.cl	DocuSigned by: Dino Moll 75EDC1EBCFB042A...
Tony Archer	AM Network	tarcher@amnetwork.com.au	DocuSigned by: Tony Archer 57D452D03BEB40D...
Tom Barber	Ernest Henry Mining	tom.barber@glencore.com.au	DocuSigned by: EEC4080ECA50437...
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